Leisure Dominates BikeShare Usage, but Commuting Trips Remain Shorter and Stable

Executive Summary

This analysis compares commuting and leisure trips' usage patterns across four U.S. metro areas, using the BikeShare database. Trips were classified as *commuting* if they occurred on weekdays during morning or evening rush hours, were made by subscribers, and were one-way. We found that leisure trips account for the majority of all bikeshare activity, but commuting trips (though fewer) are more consistent year-round and show shorter and less variable durations across all cities.

Metrics, Criteria, and Analytic Process

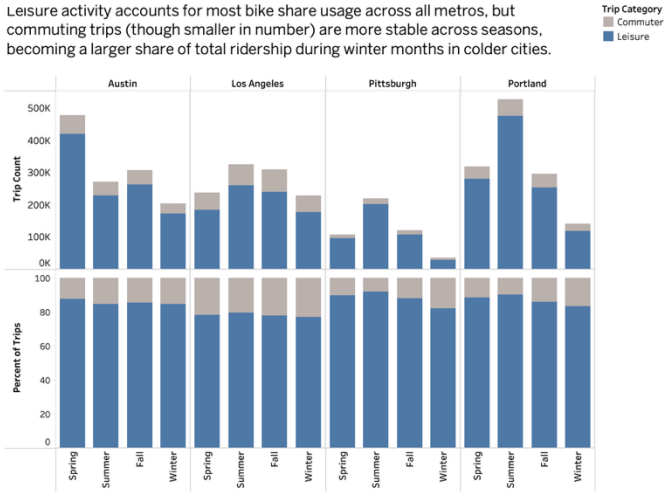
- **Trip Classification Criteria:** *Commuter trips* = Weekday (Mon–Fri), UserType = 'Subscriber', start time between 06:30-09:30 or 16:00-19:00, and one-way (FromStationID <> ToStationID). *Leisure trips* = All remaining trips.
- Metrics Calculated (by Metro and Season): TripCount (total rides by category), AvgDuration_Minutes (average trip length), DurationStdDev (variability of trip length). PercentOfTrips (Trip Count (commuter or lesiure) / Total x 100).
- Season Classification: Winter = Dec-Feb, Spring = Mar-May, Summer = Jun-Aug Fall = Sep-Nov.

Figure 1

ommuter trips are shorter and relatively more consistent across metros, while leisure rides are longer and more variable.

MetroArea	Trip Category	Avg. Duration	Std. Dev
Austin	Commuter	13.3	58.0
	Leisure	31.1	82.6
Los Angeles	Commuter	13.6	55.0
	Leisure	38.7	130.7
Pittsburgh	Commuter	30.5	141.1
	Leisure	75.6	221.4
Portland	Commuter	14.5	38.5
	Leisure	26.2	55.8

Figure 2



Key Insights and Findings

- **Insight 1:** Leisure trips dominate usage across all metros as they account for 78-90 of all rides across all metro areas (Figure 1 and 2).
- **Insight 2:** Commuting represents a smaller but more stable segment because commuter trips make up 10-22 % of rides but decline less in winter, especially in Portland and Pittsburgh, where commuter share increase to nearly 17 % (Figure 2).
- **Insight 3:** Trip durations vary significantly by purpose and city. Commuter trips are short and consistent 13-15 minutes in Austin, Los Angeles, and Portland, but much longer (approx. 30 min) in Pittsburgh due to topography and fewer nearby stations. Leisure trips vary from city to city. For example, they average 26 min in Portland but 76 min in Pittsburgh. However, lesiure trips have higher average durations and standard deviations than commuter trips for all metro areas. This indicates more flexible and unstructured travel behavior.

Appendix:

```
CREATE TEMPORARY TABLE ClassifiedTrips AS
    SELECT
        TripID,
        MetroID,
        CASE
            WHEN StartDayOfWeek BETWEEN 1 AND 5           --
Weekday
                AND UserType = 'Subscriber'               --
Regular users
                AND (
                     (StartTime BETWEEN '06:30:00' AND
'09:30:00') OR
                     (StartTime BETWEEN '16:00:00' AND
'19:00:00')
                    )
                AND FromStationID <> ToStationID          --
One-way trip
            THEN 'Commuter'
            ELSE 'Leisure'
        END AS TripCategory,
        TripDuration_Minutes,
        CASE
            WHEN MONTH(StartDate) IN (12, 1, 2) THEN 'Winter'
            WHEN MONTH(StartDate) IN (3, 4, 5) THEN 'Spring'
            WHEN MONTH(StartDate) IN (6, 7, 8) THEN 'Summer'
            WHEN MONTH(StartDate) IN (9, 10, 11) THEN 'Fall'
        END AS Season
    FROM Rentals;



SELECT
    M.CoreCity AS MetroArea,
    C.Season,
```

```
    C.TripCategory,
    COUNT(*) AS TripCount,
    AVG(C.TripDuration_Minutes) AS AvgDuration_Minutes,
    STDDEV(C.TripDuration_Minutes) AS DurationStdDev,
    ROUND(
        100.0 * COUNT(*) / SUM(COUNT(*)) OVER (PARTITION BY
M.CoreCity, C.Season),
        2
    ) AS PercentOfTrips
FROM ClassifiedTrips C
JOIN MetroArea M ON C.MetroID = M.MetroID
GROUP BY M.CoreCity, C.Season, C.TripCategory
ORDER BY M.CoreCity, FIELD(C.Season,'Spring','Summer','Fall',
'Winter'), C.TripCategory;

SELECT
    M.CoreCity AS MetroArea,
    C.TripCategory,
    COUNT(*) AS TripCount,
    AVG(C.TripDuration_Minutes) AS AvgDuration_Minutes,
    STDDEV(C.TripDuration_Minutes) AS DurationStdDev,
    ROUND(
        100.0 * COUNT(*) / SUM(COUNT(*)) OVER (PARTITION BY
M.CoreCity),
        2
    ) AS PercentOfTrips
FROM ClassifiedTrips C
JOIN MetroArea M ON C.MetroID = M.MetroID
GROUP BY M.CoreCity, C.TripCategory
ORDER BY M.CoreCity, C.TripCategory;
```